

Number Systems, Boolean Algebra & Logic Gates

A Comprehensive Overview of Positional Number Systems, Binary Codes, Axiomatic Boolean Algebra, and Fundamental Logic Realizations

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Part I: Number Systems & Conversions

Understanding radix representation, base conversions, signed binary formats, and binary codes.

Fundamentals of Number Systems

Radix & Positional Notation

A number system with radix (base) r uses r distinct digits from 0 to $r-1$.

- **Decimal (Base-10):** Digits 0–9. Standard human counting.
- **Binary (Base-2):** Digits 0, 1. Fundamental unit of digital hardware.
- **Octal (Base-8):** Digits 0–7. Shorthand grouping of 3 binary bits.
- **Hexadecimal (Base-16):** Digits 0–9, A–F. Grouping of 4 binary bits.

Signed Binary Representations

Computers represent signed numbers using specialized complement systems:

- **Sign-Magnitude:** MSB represents sign (0=+, 1=-); remaining bits give magnitude.
- **1's Complement:** Invert all bits of the positive number to get negative.
- **2's Complement:** Add 1 to the 1's complement. Simplifies arithmetic hardware (eliminates dual zero).

$$\text{2's Complement} = (\text{1's Complement}) + 1$$

Number Base Comparison Table

Decimal (r=10)	Binary 4-Bit (r=2)	Octal (r=8)	Hexadecimal (r=16)
0	0000	0	0
3	0011	3	3
7	0111	7	7
10	1010	12	A
12	1100	14	C
15	1111	17	F

Binary Codes & Data Representation

- **Binary Coded Decimal (BCD / 8421):** Encodes each decimal digit (0-9) directly into a 4-bit binary equivalent.
- **Excess-3 Code:** Unweighted code formed by adding 3 (00112) to the BCD value; simplifies 9's complement operations.
- **Gray Code:** Unit-distance non-weighted code where adjacent values differ by exactly one bit, preventing electromechanical shaft encoder errors.
- **Alphanumeric Codes:** ASCII (7-bit / 8-bit) and Unicode represent character data for digital communication.



Part II: Boolean Algebra & Simplification

Axiomatic definitions, algebraic laws, De Morgan's theorems, and
canonical SOP/POS expressions.

Core Laws of Boolean Algebra



Commutative & Associative

Order of variables does not alter truth value:

$$A + B = B + A$$

$$(A \cdot B) \cdot C = A \cdot (B \cdot C)$$



Distributive & Identity

OR distributes over AND and vice-versa:

$$A + (B \cdot C) = (A + B) \cdot (A + C)$$

$$A + 0 = A, \quad A \cdot 1 = A$$



De Morgan's Theorems

Complements of products and sums transformation:

$$A \cdot B = \overline{\overline{A} + \overline{B}}$$

$$A + B = \overline{\overline{A} \cdot \overline{B}}$$

Boolean Minimization Impact

70%

Gate Reduction Capacity

Canonical Forms & Simplification

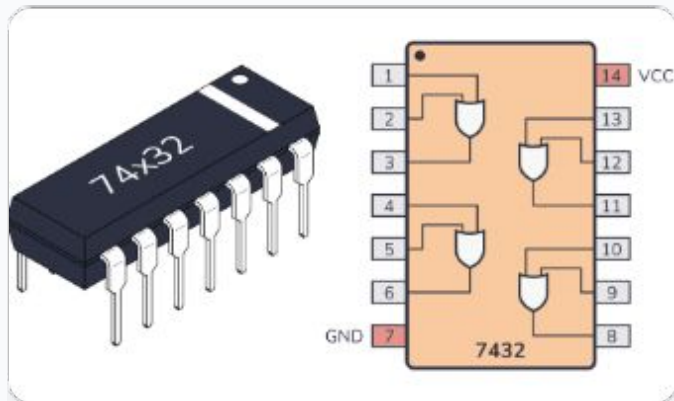
Algebraic simplification maps Boolean functions into standard canonical forms:

- **Sum of Products (SOP):** Logical OR of minterms (terms where function evaluates to 1).
- **Product of Sums (POS):** Logical AND of maxterms (terms where function evaluates to 0).
- Minimization directly minimizes total logic gate count, silicon area, and signal propagation delay in IC design.

Part III: Logic Gates & Realization

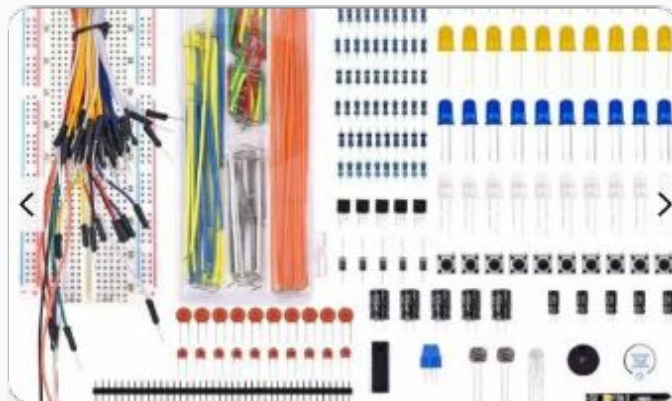
Basic, universal, and special logic gates, truth tables, and universal gate implementations.

Categorization of Logic Gates



Basic Gates

AND, OR, NOT: Foundational logic operations.
AND outputs 1 if all inputs are 1; OR outputs 1 if any input is 1.



Universal Gates

NAND, NOR: Any Boolean expression can be synthesized using exclusively NAND or exclusively NOR gates.



Special Purpose Gates

XOR, XNOR: Used extensively in binary adders, subtractors, parity checkers, and digital comparators.

Logic Minimization Workflow



Questions & Discussion

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